Encryption Scheme

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1 Yao's Garbled Circuits

2 Fairplay

[1]

1. Encryption:
   (a) Generate pseudo-random pads: SHA1($w_i \parallel k \parallel x \parallel y$) and SHA1($w_j \parallel k \parallel x \parallel y$), w_i, w_j input wire labels, k gate index, x, y input bits. Each pad is a 160-bit hash value.
   (b) Apply XOR: $ciphertext = output_label \oplus pad_i \oplus pad_j$.
2. Decryption: $output_label = ciphertext \oplus pad_i \oplus pad_j$.
3. Pros:
   (a) SHA-1 supports variable-length inputs (AES uses block ciphers).
   (b) Wire labels can be used to generate pads, without introducing additional keys.
   (c) XOR is simple and efficient.
4. Cons:
   (a) SHA-1 is outdated.
   (b) Existing weakness: when $w_i = w_j \rightarrow pad_i = pad_j$, $pad_i \oplus pad_j = 0$. [2]
   (c) Uses SHA-1 as a pseudo-random function.

3 Proof of Security

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1. Encryption:
   (a) Double encryption: $C_{x,y} = E_{k_1^x}\left(E_{k_2^y}(k_3^{g(x,y)})\right)$, k_1^x, k_2^y input wire labels, $k_3^{g(x,y)}$ output label, E symmetric encryption.
   (b) $E_k(x) = \langle r,\ f_k(r) \oplus x0^n \rangle$, r random string (n-bits), x plain text (n-bits).
2. Note: requirements for secure encryption scheme: CPA security, secure for multiple messages, elusive range, efficiently verifiable range, and double-encryption. security.

References

[1] Dahlia Malkhi, Noam Nisan, Benny Pinkas, and Yaron Sella. Fairplay—a secure two-party computation system. In *Proceedings of the 13th USENIX Security Symposium*, 2004.

[2] Raine Nieminen and Thomas Schneider. Breaking and fixing garbled circuits when a gate has duplicate input wires. *Journal of Cryptology*, 36(34), 2023.